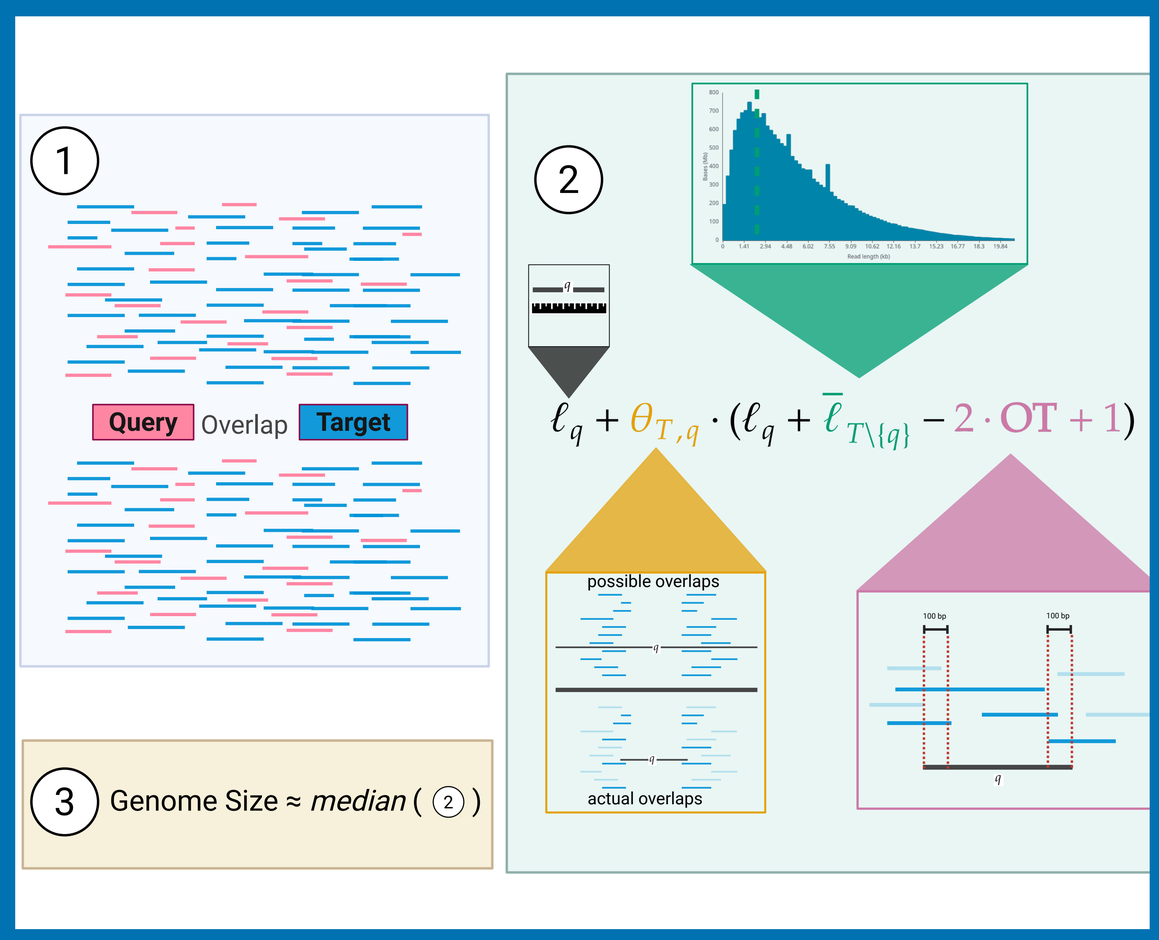
How big is your genome?



Genome size estimation from long read overlaps

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BACKGROUND

- **Estimating genome size** is important for understanding genome structure, determining sequencing coverage, and guiding downstream analyses such as assembly.
- Most tools use **short-read data** and rely on **k-mer profiles**, which don't work well with **long-read technologies**.
- We developed **LRGE**, a fast and accurate genome size estimator that works on long reads using **read-to-read overlaps**.

METHODS

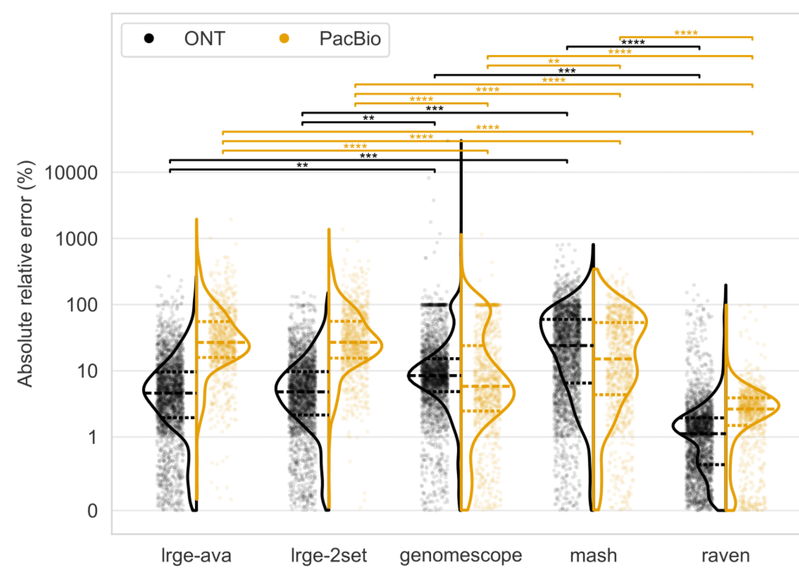
Key points:

- Estimate genome size by analysing **how many reads overlap each other**.
- Reads with many overlaps suggest a **smaller genome**; fewer overlaps a **larger genome**.
- The **median** of these per-read estimates gives a genome size estimate with 92% CI.

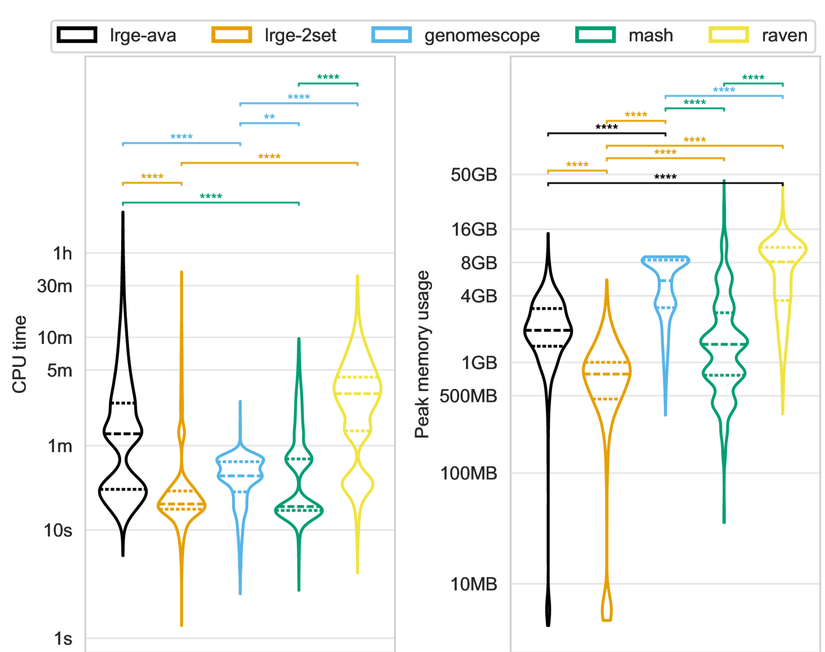
Tools & Implementation:

- Implemented in **Rust**, using **minimap2 FFI bindings** for efficient overlap detection.
- Two modes:
  - 2set: overlaps between small query and large target set.
  - ava: all-v-all overlaps.

RESULTS



Absolute relative error (y-axis) for each method's (x-axis) genome size estimation on ONT and PacBio data. The y-axis is scaled according to a symmetric logarithm, which is linear between -1 and 1 and logarithmic (base 10) thereafter. The statistical annotations are the result of a Tukey's range test and are coloured by the sequencing platform being compared. Pairs with no annotation indicate no significant difference. The dashed lines in the violins are the quartiles.

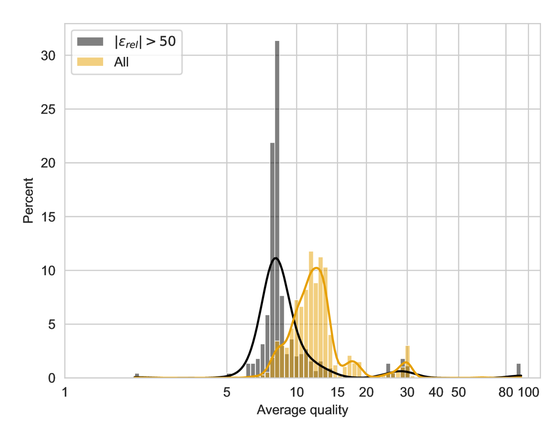


CPU time (left y-axis) and maximum memory usage (right y-axis) for each method (colours). The y-axis is log-scaled (base 10). The statistical annotations are the result of a Tukey's range test and are coloured by the method with the lower mean value. Pairs with no annotation indicate no significant difference. The dashed lines in the violins are the quartiles.

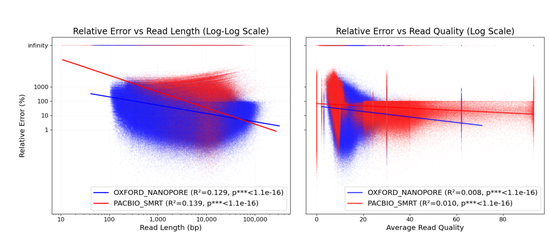
CONCLUSION

- LRGE is a **lightweight**, **fast** tool for genome size estimation from long read data.
- Outperforms k-mer-based tools, with **lower computational cost** than assembly.
- Useful for assembly and estimating coverage.

Recommendation: Filter out low-quality and very short reads to improve estimates.



Distribution of average read quality for samples with a relative error greater than 50% and all other samples.



Impact of read length and quality on LRGE estimation accuracy. Lines are a linear regression fit to the points with the significance of the fit and the coefficient of determination (R^2) in the legend. Infinite estimates are those reads which have no overlaps.

mbhall88/lrge

ACKNOWLEDGEMENTS:

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Poster

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